CPE 314 — Software Design

University of Cebu

Computer Engineering Department

UC-B Campus Marketplace with Reviews

Project Design Document

Table 1: Group information

Group Name / No.:	Group 1 / Team 01 | BSCPE 3B | EDP 69323
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Revision History

Table 2: Document revisions

Version	Date	Author	Changes
0.1	17 Oct 2026	Group 1	Initial planning draft.
0.2	18 Oct 2026	Group 1	Working title, roster and assigned theme updated.
0.3	23 Oct 2026	Group 1	Week 8 template draft with introduction, ER sketch and roles.

Formatting Guidelines

Maintain these template rules as the same document develops through the later milestones.

Table 3: Formatting rules from the template

Element	Rule
Body font	Calibri, 11 pt.
Headings	Use Word Heading 1, Heading 2 and Heading 3 styles so the table of contents can update.
Margins	One inch on all sides.
Line spacing	Body: 1.15–1.5. Tables: single spacing.
Table captions	Table N: short description, directly above each table.
Figure/chart captions	Figure N: short description, directly below each figure.
Code snippets	Consolas or Courier New, 9–10 pt; fewer than about 20 lines; link to full repository files.
Page numbers	Use automatic page-number fields in the footer.
Alignment	Body text left aligned or justified; headings left aligned.

1. Introduction & Project Overview

UC-B Campus Marketplace with Reviews is a proposed Java desktop application for students, faculty, school units, and department committees to post and browse campus goods. Our assigned theme is Option J, retaining Textbook, Electronics, and Furniture listing types with different condition-grading rules, nested category search, buyer-interest waitlists, recently viewed listings, and reviews. We plan to use JavaFX, SQLite through JDBC, a background sales-history CSV import, FakeStoreAPI product suggestions, and category price and rating bar charts. Uniforms and department merchandise are proposed additions for instructor review; the initial application uses local storage, with online payments outside its scope.

2. Preliminary ER Sketch & Team Roles

The preliminary model carries forward the entities, key attributes and relationships from the proposal. The sketch below shows the core listing relationships; Table 4 lists all seven proposed records and their foreign keys.

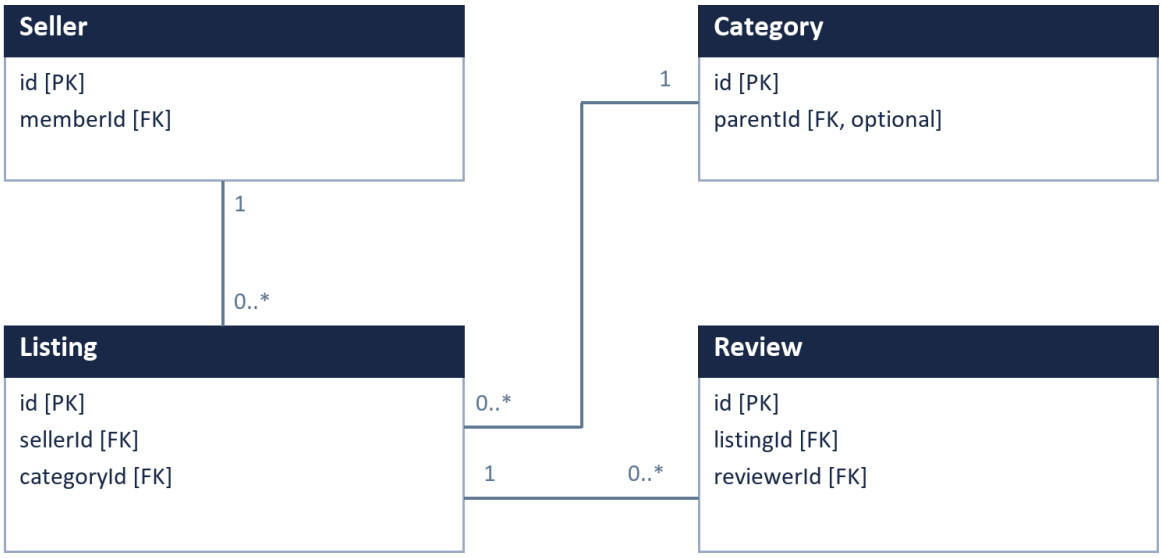


Figure 1: Core preliminary entity relationships

Each Seller and Category can relate to many Listings; each Listing can receive many Reviews. The 1 and 0..* labels mean one parent and zero or more related records. This is a proposal sketch; the final ERD and actual schema will be documented in Section 4.

2.1 Key Attributes and Relationships

PK identifies a record uniquely. FK refers to a related record. Category.parentId may be empty for a root category.

Table 4: Preliminary entities and key attributes

Entity	Key attributes
Member	id PK; name
Seller	id PK; memberId FK → Member.id; displayName; sellerKind
Category	id PK; parentId FK → Category.id; name
Listing	id PK; sellerId FK → Seller.id; categoryId FK → Category.id; type; title; priceMinor; currency; status
Review	id PK; listingId FK → Listing.id; reviewerId FK → Member.id; rating; comment
BuyerInterest	id PK; listingId FK → Listing.id; buyerId FK → Member.id; joiningOrder
HistoricalSale	id PK; categoryId FK → Category.id; sourceRecordId; priceMinor; currency; soldAt

Member relates to many Seller profiles, Reviews and BuyerInterest entries. Category relates to many Listings, child Categories and HistoricalSale records. Listing relates to many Reviews and BuyerInterest entries. Each listing is proposed to have one matching subtype-details record; its fields are not finalized.

2.2 Team Roles

Table 5: Proposed primary responsibilities for Week 8

Member	Primary responsibility
John Benedict Alaiza	Sample listings, categories and condition examples.
Acexion B. Mahigugmaon	Architecture, implementation planning and integration.
Glyde V. Arreglado	ER sketch, attributes and relationship review.
Tristan Dominic Urot	GUI sketches, control labels and user steps.
Joshua Brian S. Enovero	Sample CSV and API/import validation cases.
Eman De Paz	Test cases and expected analytics calculations.

These are proposed responsibilities for member confirmation. Actual contributions and supporting commits will be recorded in Section 9.

3. System Architecture & Object-Oriented Design

Reserved for a later milestone. Add the class hierarchy, responsibilities and design rationale after the proposal is approved and the implementation is developed.

4. Data Model & Database Design

Reserved for a later milestone. Finalize the ERD, SQL types, constraints and SQLite schema. Section 2 contains the preliminary proposal model only.

5. API Integration Design

Reserved for a later milestone. Document the actual endpoint response, parsing, comparable-product matching, currency policy and failure handling after verification.

6. Concurrency Design

Reserved for a later milestone. Record the implemented background import/API work, UI update approach, cancellation and error handling.

7. Data Mining & Visualization Design

Reserved for a later milestone. Document the implemented category calculations, sample counts, currency handling and labeled charts.

8. Testing Summary

Reserved for a later milestone. No application test suite or execution results are recorded in this Week 8 draft. Add actual JUnit coverage, edge cases and results when available.

9. Individual Contributions

Contribution records will be added with verified commit references. The roles in Section 2 describe proposed responsibilities and do not claim completed implementation.

Appendix

This Week 8 draft contains the title page, written overview, preliminary ER sketch and six-member role list. Later sections will be completed in this same cumulative document as evidence becomes available.

Project References

Project repository
https://github.com/XionAce/cpe314-69323-capstone-team-01

Project proposal
https://github.com/XionAce/cpe314-69323-capstone-team-01/blob/project/proposal/docs/proposal.md

Student Guide — Group Capstone Project
https://drive.google.com/file/d/195Y8cNaHnd0M7FpIwmKDYzfok5pbKZti/view

Week 8 design-document checklist
https://drive.google.com/file/d/1wb9xb2nd7RDowEEbSMx-M-l58qnO9gMM/view

Project design-document template
https://drive.google.com/file/d/1sRJ5rjciHBeYgBQmUbpJB12kWyD_sCkO/view

FakeStoreAPI documentation
https://fakestoreapi.com/docs

Document Status

Instructor approval, final implementation ownership, API behavior and currency decisions remain to be confirmed. This draft contains no claim of a working application or completed tests.